

Real-time GS Mapping on Aria

Building useful maps under limited compute

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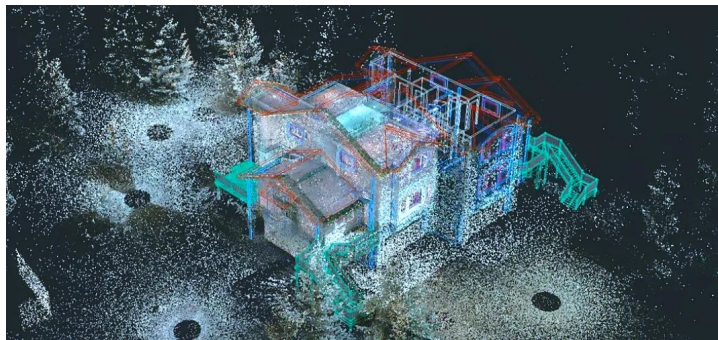
RPM Lab, Seoul National University

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Why Gaussian Splatting?

Represent a scene with colored, semi-transparent 3D Gaussians.

Point cloud



Gaussian representation / rendering



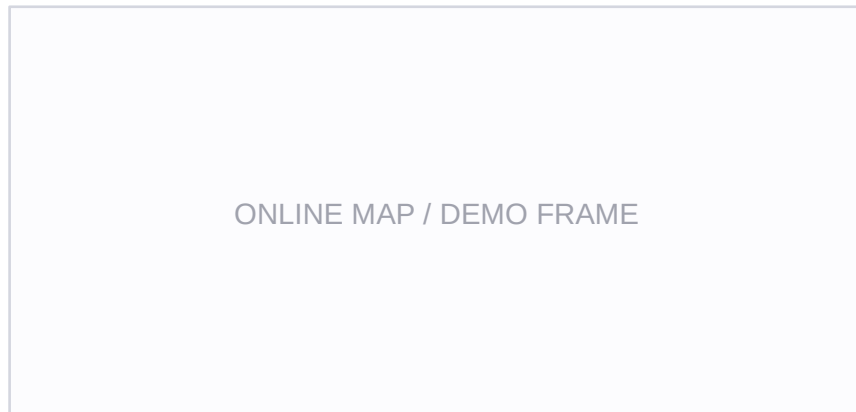
Photorealistic appearance + fast rendering

Why Real-time Mapping?

A robot or AR user needs a useful map while moving.



Observe the environment

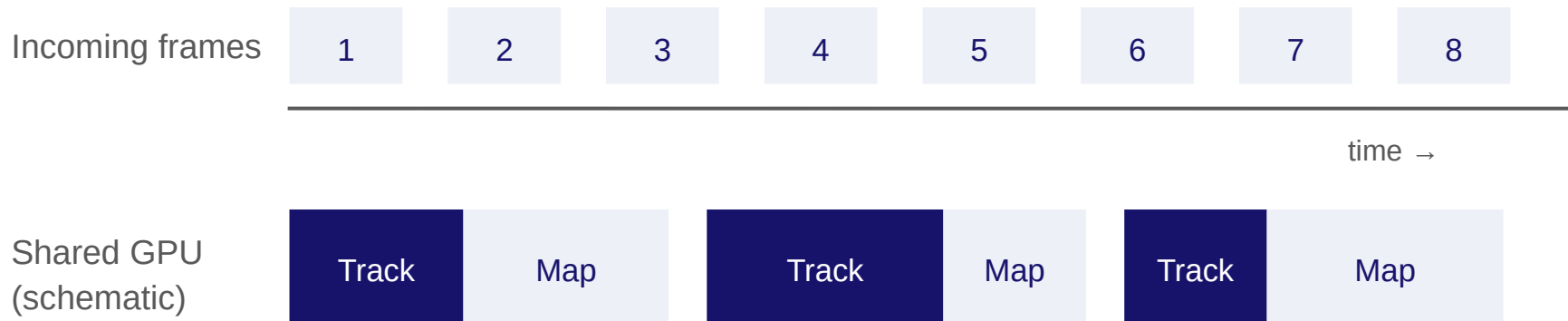


Build and use the map now

Fast rendering does not guarantee fast map convergence.

Mapping under Limited Compute

Tracking and mapping share a finite online compute budget.



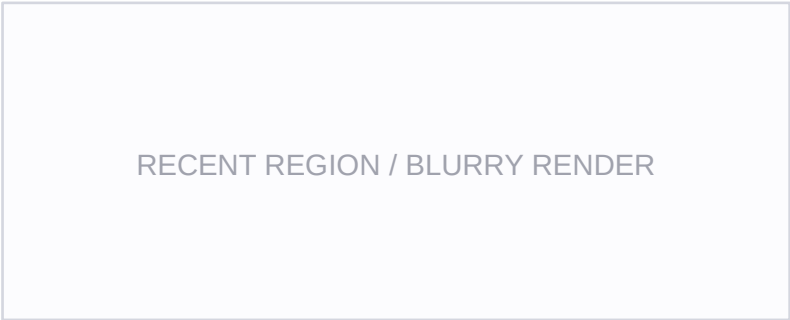
More mapping views compete for the same updates.

Use available observations within available computation.

What Makes Online Mapping Difficult?

Two requirements: convincing appearance and reliable geometry.

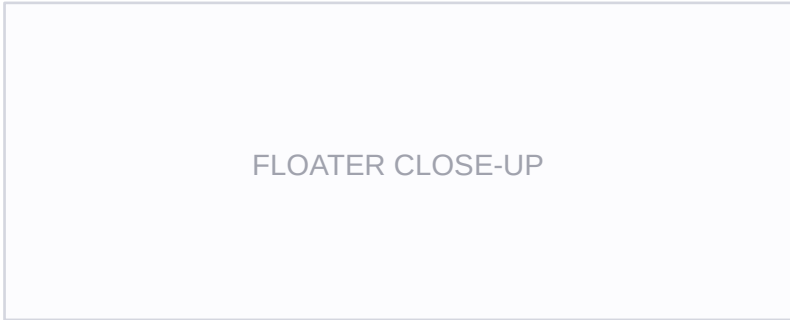
Under-optimized appearance



RECENT REGION / BLURRY RENDER

Too little useful supervision

Incorrect geometry



FLOATER CLOSE-UP

Opacity in observed free space

How quickly can a newly observed region become reliable?

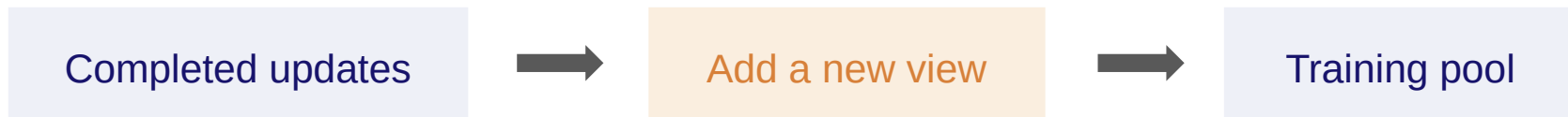
When Should We Add More Views?

Use frames between keyframes, paced by completed optimization.

Tracking
keyframes



Intermediate RGB frames also carry mapping information.



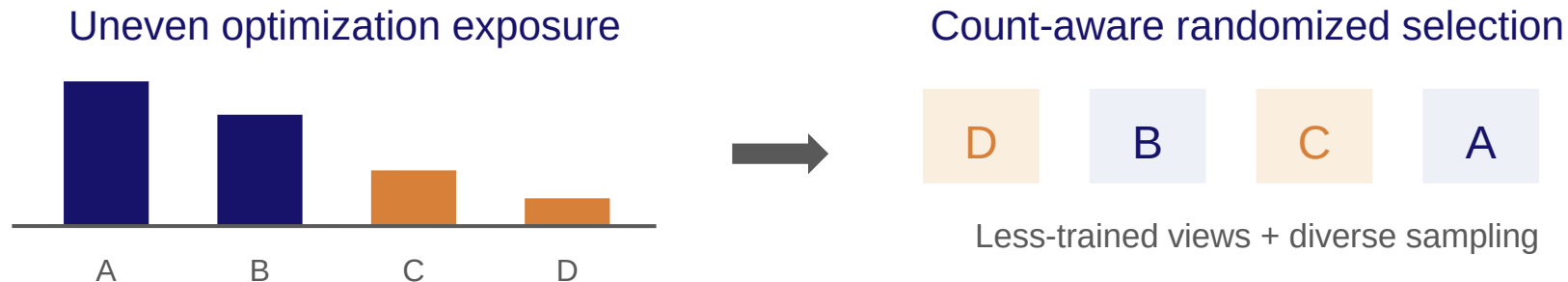
$$N(t) - N_0 \approx \frac{S(t)}{\kappa}$$

N: training-pool size S: completed GPU mapping updates κ : updates per added view

Grow supervision at the pace of optimization.

Which Views Should We Optimize?

Give under-trained observations more opportunities, while mixing views.



Normalized variance
+ entropy

$$\min_{p \in \Delta_N} \mathbb{E}_{i \sim p} [\Delta_i V_{\text{norm}}(\mathbf{n})] - \tau H(p)$$

Softmax solution

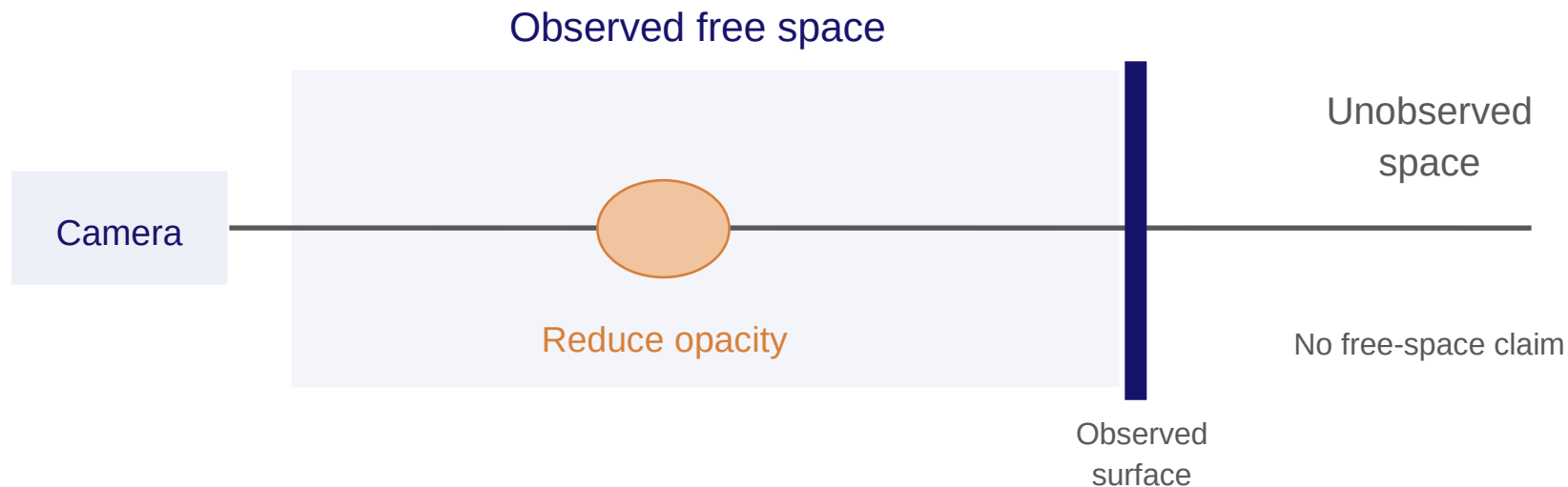
$$p_i^* = \text{softmax}_i(-\beta n_i) = \frac{e^{-\beta n_i}}{\sum_j e^{-\beta n_j}}$$

$$V_{\text{norm}}(\mathbf{n}) = \frac{1}{2N} \sum_i (n_i - \bar{n})^2$$

n_i : selection count H : sampling entropy

Keeping Observed Free Space Empty

A good color match can still hide an incorrect 3D structure.

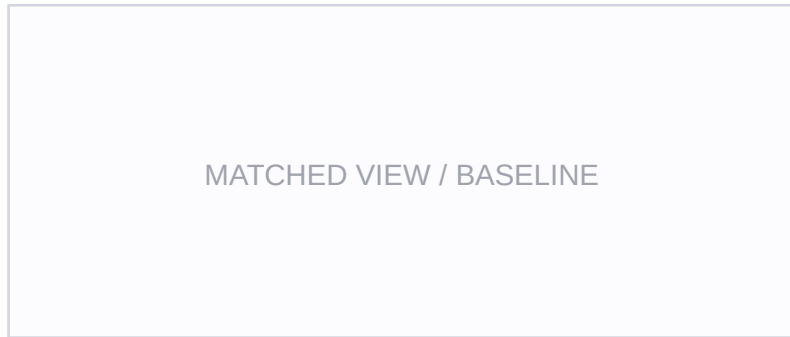


Use causal depth evidence to suppress premature opacity.

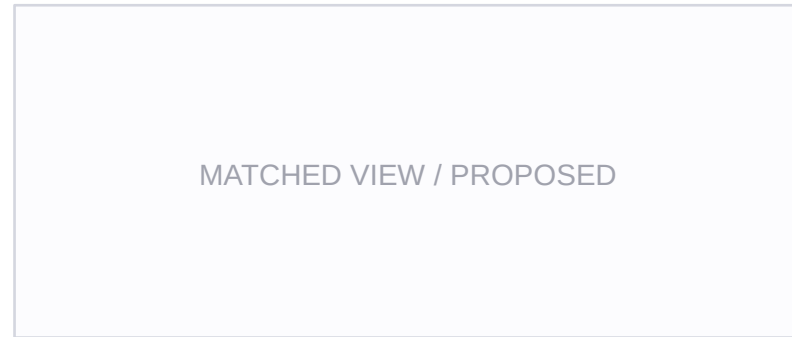
Current Progress & Evaluation

View growth and scheduling are implemented; joint validation is ongoing.

Baseline



Proposed setting



Next: matched-budget appearance + geometry evaluation

Takeaway & Next Step

Make each unit of online computation
produce a more useful map.

01

When to add

Pace view growth

02

What to optimize

Balance exposure

03

Where to constrain

Use free-space evidence

Next: validate appearance and geometry gains at the native input rate.